

Spherical-Shell Stokes Flow Benchmark Using Underworld3

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Abstract. We reproduce the analytical spherical-shell Stokes benchmark of Thieulot (2017) (hereafter T2017) using Underworld3 (hereafter UW3; Moresi et al., 2025). The benchmark describes incompressible viscous Stokes flow in a spherical shell with radially directed gravity, tangential velocity boundary conditions, and a power-law radial viscosity. We consider the two reference cases of T2017: $m = -1$, for which the viscosity is constant, and $m = 3$, for which the viscosity varies radially across the shell. The UW3 calculations use a $P_2 \times P_1$ Taylor–Hood finite-element discretisation on progressively refined spherical-shell meshes. Velocity and pressure accuracy is evaluated using relative volume L_2 -norm errors, while radial normal stress and pressure on the inner and outer spherical boundaries are evaluated using boundary L_2 -norm measures. The calculations recover the expected smooth-solution convergence hierarchy, with approximately third-order velocity convergence and second-order pressure convergence. Boundary radial normal-stress and boundary pressure errors also approach second-order convergence at the finest resolutions. These results verify the UW3 spherical-shell Stokes implementation and provide a compact test for velocity, pressure, and boundary-stress post-processing in curved three-dimensional geometry.

1 Introduction

Accurate solution of the incompressible Stokes equations is central to global mantle-convection and planetary-interior modelling. Spherical-shell calculations are more representative of global mantle flow than Cartesian boxes, but they also test numerical implementations more strongly. The discretisation must handle curved inner and outer boundaries, radial gravity, pressure gauge freedom, radial viscosity variation, tangential boundary conditions, and stress quantities evaluated on non-planar surfaces.

Analytical benchmarks provide a direct way to measure discretisation error. Unlike code-to-code comparisons, they supply a fixed reference velocity and pressure field, so convergence rates can be evaluated under systematic mesh refinement. This is especially useful for geodynamic Stokes solvers because the same test exercises velocity–pressure coupling, incompressibility, boundary condition implementation, pressure normalisation, curved-domain integration, and stress recovery.

The spherical-shell benchmark introduced by T2017 provides a closed-form solution for viscous incompressible Stokes flow with radially directed gravity and power-law radial viscosity. The solution is axisymmetric, has no normal flow on the inner and outer boundaries, and supplies analytical expressions for density, velocity, pressure, body force, and the full stress tensor. Related analytical shell-flow solutions and finite-element verification studies have been developed by Kramer et al. (2021) and

Davies et al. (2022), while boundary normal stress has long been used as a diagnostic in spherical geodynamics benchmarks (Zhong et al., 1993, 2008; Liu and King, 2019).

In this study, we use T2017 to verify the spherical-shell Stokes implementation in UW3. The article documents the governing equations, analytical benchmark structure, numerical setup, error metrics, and observed convergence of velocity, pressure, boundary pressure, and radial normal stress. The focus is on whether a $P_2 \times P_1$ Taylor–Hood discretisation recovers the expected smooth-solution convergence hierarchy in curved three-dimensional geometry.

2 Governing Equations and Benchmark Solution

The benchmark solves the steady incompressible Stokes equations

$$\nabla \cdot \mathbf{u} = 0, \tag{1a}$$

$$35 \quad \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g} = \mathbf{0}, \tag{1b}$$

where $\mathbf{u}$ is velocity, ρ is density, and $\mathbf{g} = -\mathbf{e}_r$ is radial gravity of unit magnitude. The stress is

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\mu(r)\boldsymbol{\varepsilon}(\mathbf{u}), \tag{2}$$

with pressure p , viscosity μ , and strain-rate tensor

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^T). \tag{3}$$

40 Following T2017, spherical coordinates (r, θ, ϕ) are used, with θ the polar angle and ϕ the azimuth. All fields are independent of ϕ . The tangential velocity components are prescribed in the separable form

$$u_\theta(r, \theta) = f(r) \sin \theta, \tag{4a}$$

$$u_\phi(r, \theta) = f(r) \sin \theta, \tag{4b}$$

and incompressibility gives the radial velocity

$$45 \quad u_r(r, \theta) = g(r) \cos \theta, \quad g(r) = -\frac{2}{r^2} \int r f(r) dr. \tag{5}$$

The boundary condition $u_r = 0$ is imposed on the inner and outer shell surfaces $r = R_1$ and $r = R_2$, so the flow is tangential on both boundaries. The viscosity is

$$\mu(r) = \mu_0 r^{m+1}, \tag{6}$$

where $m = -1$ gives constant viscosity and $m = 3$ gives the radially varying case considered here.

50 The pressure has the form

$$p(r, \theta) = h(r) \cos \theta, \tag{7}$$

where the radial pressure coefficient is

$$h(r) = \frac{2\mu_0}{r}g(r) \quad \text{for } m = -1, \quad h(r) = \frac{m+3}{r}\mu(r)g(r) \quad \text{for } m \neq -1. \quad (8)$$

For $m \neq -1$, the radial velocity function is obtained from

$$55 \quad f(r) = \alpha r^{-(m+3)} + \beta r, \quad (9)$$

with α , β , and the integration constant in $g(r)$ chosen so that $g(R_1) = g(R_2) = 0$. The $m = -1$ case has the corresponding logarithmic radial function given by T2017. In both cases the density follows from the radial Stokes balance after substituting the analytical velocity, pressure, viscosity, and gravity fields. This construction yields the closed-form density, velocity, pressure, body force, and stress tensor used for the UW3 error calculations.

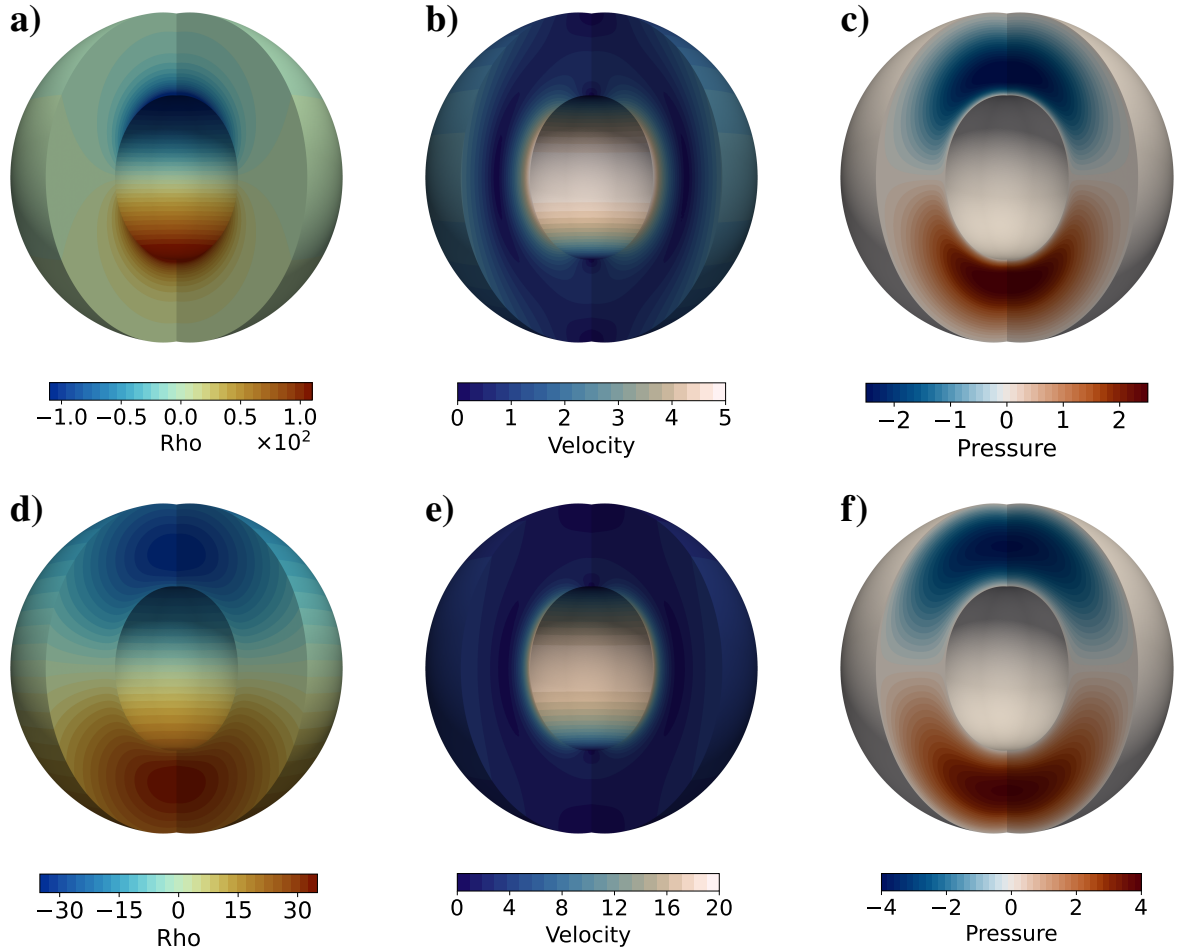


Figure 1. Analytical density, velocity magnitude, and pressure fields. The upper row shows the $m = -1$ case, and the lower row shows the $m = 3$ case.

60 Figure 1 illustrates the analytical fields used for verification. Both cases satisfy the zero-normal-flow condition on the inner and outer boundaries while retaining non-zero tangential velocity. The $m = 3$ case uses the same shell geometry but introduces stronger radial viscosity variation, making pressure and stress recovery more demanding than in the constant-viscosity case.

3 Numerical Setup

The benchmark is implemented in UW3, a finite-element geodynamics framework built on PETSc (Balay et al., 2024) with symbolic problem specification in Python. The spherical-shell meshes are generated through the Gmsh meshing backend (Geuzaine and Remacle, 2009). The reported calculations use a $P_2 \times P_1$ Taylor–Hood element pair, with quadratic velocity and continuous linear pressure. For sufficiently smooth Stokes solutions, this mixed pair is expected to give third-order velocity convergence and second-order pressure convergence in volume L_2 norms (Girault and Raviart, 1986; Boffi et al., 2013).

70 Mesh resolution is reported using the characteristic cell size $h = 1/8, \dots, 1/128$. The analytical velocity is imposed on both spherical boundaries. The pressure is determined only up to an additive constant, so the numerical and analytical pressure fields are compared after applying the same mean-pressure gauge.

4 Error Metrics

For a numerical quantity q_h and analytical field q^* , the relative volume error is

$$E_{L_2}^{\text{rel}}(q) = \left(\frac{\int_{\Omega} |q_h - q^*|^2 d\Omega}{\int_{\Omega} |q^*|^2 d\Omega} \right)^{1/2}. \quad (10)$$

75 This metric is used for velocity and pressure. Boundary diagnostics are evaluated on the inner and outer spherical surfaces, denoted Γ_{inner} and Γ_{outer} , using

$$E_{L_2, \Gamma}^{\text{rel}}(q) = \left(\frac{\int_{\Gamma} |q_h - q^*|^2 d\Gamma}{\int_{\Gamma} |q^*|^2 d\Gamma} \right)^{1/2}. \quad (11)$$

Absolute boundary pressure errors use the corresponding unnormalised measure

$$E_{L_2, \Gamma}(p) = \left(\int_{\Gamma} |p_h - p^*|^2 d\Gamma \right)^{1/2}. \quad (12)$$

80 Observed convergence rates between two consecutive mesh sizes h_1 and h_2 are computed as

$$r = \frac{\log(E_{h_1}/E_{h_2})}{\log(h_1/h_2)}. \quad (13)$$

5 Results

5.1 Velocity and pressure

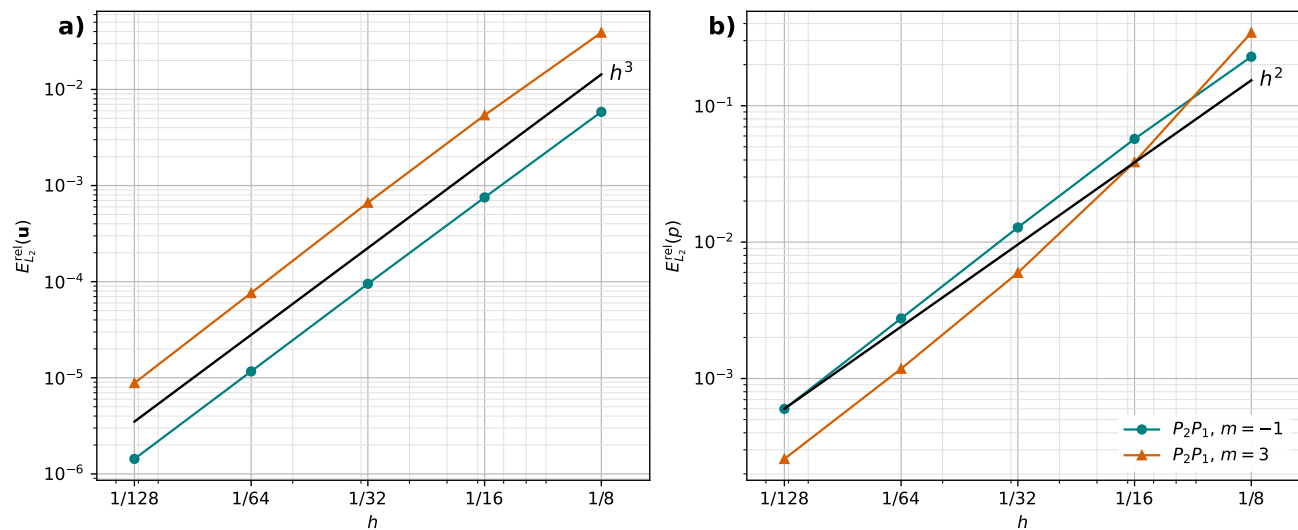


Figure 2. Convergence of the volume L_2 -norm velocity and pressure errors for the $P_2 \times P_1$ discretisation.

Table 1. Spherical-shell T2017 benchmark convergence for $m = -1$ and $m = 3$.

h	$m = -1$				$m = 3$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	velocity rate	$E_{L_2}^{\text{rel}}(p)$	pressure rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	velocity rate	$E_{L_2}^{\text{rel}}(p)$	pressure rate
1/8	5.842e-03	–	2.287e-01	–	3.899e-02	–	3.433e-01	–
1/16	7.521e-04	2.96	5.709e-02	2.00	5.391e-03	2.85	3.858e-02	3.15
1/32	9.500e-05	2.98	1.281e-02	2.16	6.630e-04	3.02	5.959e-03	2.69
1/64	1.166e-05	3.03	2.757e-03	2.22	7.628e-05	3.12	1.179e-03	2.34
1/128	1.432e-06	3.03	5.986e-04	2.20	8.816e-06	3.11	2.573e-04	2.20

The velocity and pressure errors are shown in Fig. 2 and Table 1. The velocity error approaches $O(h^3)$ for both $m = -1$ and $m = 3$, while the pressure error approaches $O(h^2)$ at the finest resolutions. The pressure rates are less regular on the coarsest meshes, particularly for the variable-viscosity case, but the refined-mesh trend is consistent with the expected Taylor–Hood hierarchy and with the convergence behaviour reported by T2017.

5.2 Boundary Stress and Pressure

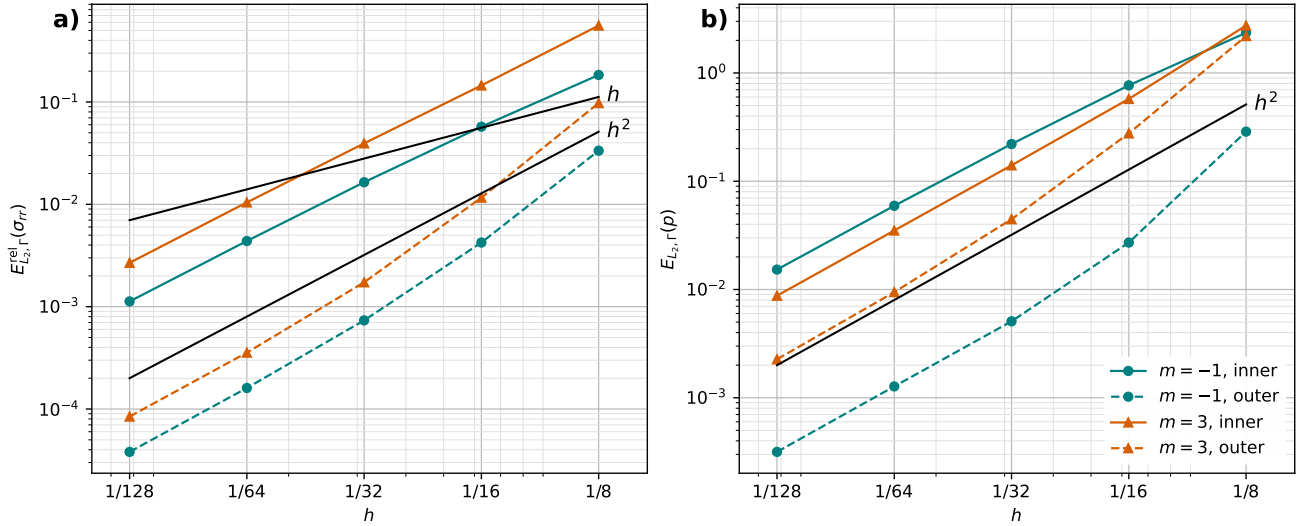


Figure 3. Convergence of boundary radial normal-stress and pressure errors. Solid lines denote the inner boundary and dashed lines denote the outer boundary.

Table 2. Spherical-shell T2017 benchmark convergence for boundary σ_{rr} errors.

h	$m = -1$				$m = 3$			
	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	inner rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	outer rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	inner rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	outer rate
1/8	1.836e-01	–	3.343e-02	–	5.578e-01	–	9.755e-02	–
1/16	5.732e-02	1.68	4.233e-03	2.98	1.450e-01	1.94	1.162e-02	3.07
1/32	1.644e-02	1.80	7.347e-04	2.53	3.941e-02	1.88	1.730e-03	2.75
1/64	4.382e-03	1.91	1.605e-04	2.19	1.044e-02	1.92	3.556e-04	2.28
1/128	1.126e-03	1.96	3.811e-05	2.07	2.684e-03	1.96	8.435e-05	2.08

Table 3. Spherical-shell T2017 benchmark convergence for absolute boundary pressure errors.

h	$m = -1$				$m = 3$			
	$E_{L_2, \Gamma_{\text{inner}}}(p)$	inner rate	$E_{L_2, \Gamma_{\text{outer}}}(p)$	outer rate	$E_{L_2, \Gamma_{\text{inner}}}(p)$	inner rate	$E_{L_2, \Gamma_{\text{outer}}}(p)$	outer rate
1/8	2.351e+00	–	2.870e-01	–	2.739e+00	–	2.192e+00	–
1/16	7.683e-01	1.61	2.712e-02	3.40	5.756e-01	2.25	2.766e-01	2.99
1/32	2.206e-01	1.80	5.082e-03	2.42	1.403e-01	2.04	4.455e-02	2.63
1/64	5.933e-02	1.89	1.273e-03	2.00	3.505e-02	2.00	9.456e-03	2.24
1/128	1.528e-02	1.96	3.162e-04	2.01	8.757e-03	2.00	2.267e-03	2.06

Although T2017 derive the full stress tensor, the published benchmark focuses primarily on velocity and pressure convergence. Here the radial normal stress, σ_{rr} , is included as an additional diagnostic because normal stress on spherical boundaries is important for dynamic topography and surface-loading calculations. Figure 3 and Tables 2–3 show that σ_{rr} and boundary pressure errors approach approximately second-order convergence on both shell boundaries for both viscosity cases. The inner-boundary error constants are generally larger than the outer-boundary constants, but the asymptotic slopes are consistent across the two surfaces.

6 Discussion

The results verify three aspects of the UW3 spherical-shell Stokes implementation. First, the $P_2 \times P_1$ discretisation recovers the smooth-solution hierarchy: velocity converges one order faster than pressure in the L_2 norm. Second, comparable rates are obtained for both constant and radially variable viscosity, although the $m = 3$ case has larger error constants and stronger pre-asymptotic pressure behaviour. Third, the boundary radial normal-stress diagnostic converges at roughly second order, consistent with related spherical-shell stress verification studies (Kramer et al., 2021).

The boundary error magnitudes differ between the inner and outer surfaces. This does not indicate a failure of the benchmark. The relative norms are normalised separately on two surfaces with different radii, analytical stress amplitudes, curvatures, and local combinations of pressure and deviatoric-stress error. The absolute pressure errors show the same behaviour because the pressure trace is sampled on geometrically distinct surfaces. The relevant verification criterion is therefore the asymptotic slope, not equality of the error constants.

The benchmark is deliberately compact but still probes several practical issues for spherical geodynamics solvers: curved boundary representation, pressure normalisation, radial viscosity variation, tangential velocity boundary conditions, and stress recovery on spherical surfaces. These properties make it a useful regression test alongside annulus and Cartesian Stokes benchmarks.

7 Conclusions

We have presented a verification study of the T2017 spherical-shell Stokes benchmark using UW3. The benchmark uses analytical velocity, pressure, density, viscosity, and stress fields in a curved three-dimensional shell. For the $P_2 \times P_1$ Taylor–Hood discretisation, UW3 recovers approximately $O(h^3)$ velocity convergence and $O(h^2)$ pressure convergence for both the $m = -1$ and $m = 3$ cases. Boundary radial normal-stress and boundary pressure diagnostics also approach second-order convergence at the finest resolutions. These results support the use of the T2017 spherical-shell benchmark as a production verification case for UW3 velocity, pressure, and boundary-stress post-processing.

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